

Lab 4.2.6 Report: Working with Text Files in the CLI

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Part 1: Graphical Text Editors

Tasks Completed:

- **Opened SciTE from the GUI**
- **Created a file named space.txt with the given content**
- **Saved the file in the home directory**
- **Reopened the file**
- **Opened SciTE from the Terminal**

Questions and Answers:

Question 1: Could you immediately find space.txt?

Answer: No, I could not immediately find the space.txt file because:

- **SciTE searches only for file types it recognizes by default**
- **The .txt extension is not recognized by default**
- **Solution: Changed the file type dropdown to "All Files (*)" to display all files**
- **This shows that while Linux file systems don't rely on extensions, some GUI applications use them to identify file types**

Question 2: Why is the prompt not shown in the terminal?

Answer:

- **When you launch a program from the terminal, it runs in the foreground**
 - **The terminal remains in a "waiting state" until the program terminates**
 - **The prompt reappears when the program closes**
 - **Alternative: You can press Ctrl+C to terminate the process and return to the prompt immediately**
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Part 2: Command Line Text Editors (Nano)

Tasks Completed:

- **Opened space.txt with nano editor**
- **Navigated through the file using Home and End keys**
- **Explored the nano help menu (Ctrl+G)**
- **Learned nano keyboard shortcuts**

Questions and Answers:

Question 3: What character does nano use to represent that a line continues beyond the boundaries of the screen?

Answer:

- **Nano uses the \$ (dollar sign) character to indicate line continuation**
 - **This symbol appears at the end of lines that extend beyond the screen boundaries**
 - **Users can navigate using Home and End keys to reach the beginning and end of long lines**
 - **This is a helpful visual indicator for text that doesn't fit on the current display**
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Part 3: Working with Configuration Files

Section 1: Locating Configuration Files

Tasks Completed:

- **Located user configuration files in the home directory**
- **Found hidden configuration files using ls -la**
- **Viewed system-wide configuration files in /etc**
- **Examined the .bashrc file content**

Key Findings:

User Configuration Files:

- **Location: Home directory (/home/analyst or ~)**

- **Naming Convention:** Hidden files starting with . (dot)
- **Examples:** .bashrc, .bash_profile, .bashrc_stock
- **Permission:** Users have full write access to their home directory

System Configuration Files:

- **Location:** /etc directory
- **Examples:** nginx, bash.bashrc, ssh, sudoers
- **Permission:** Only root users can modify these files
- **Purpose:** Control system-wide services and applications

Question 4: Why are user application configuration files saved in the user's home directory and not under /etc with all the other system-wide configuration files?

Answer:

- **User configuration files are personal to one user only, while /etc files affect the entire system**
 - **Regular users have write permissions to their home directories but NOT to /etc**
 - **This permission structure provides security and prevents non-root users from modifying system-wide settings**
 - **Users can customize their applications independently without affecting other users**
 - **Only root can modify /etc files, maintaining system integrity and security**
 - **This separation of concerns follows Linux security best practices**
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Section 2: Editing and Saving Configuration Files

Tasks Completed:

- **Edited .bashrc file using both SciTE and Nano**
- **Changed prompt color from green to red using SciTE**
- **Changed prompt color from red to yellow using Nano**
- **Verified changes by opening new terminal windows**

Color Codes Used:

- 31 = Red
- 32 = Green
- 33 = Yellow

Question 5: Did the terminal window which was already open also change color from green to red? Explain.

Answer: No, the terminal window that was already open did NOT change color.

Explanation:

- Configuration files are loaded when a new terminal session is started
 - The old terminal instance was already running with the previous configuration
 - It continues to use the green color that was set when it originally opened
 - Changes to .bashrc only affect newly opened terminal windows
 - Solution: Either open a new terminal or execute the bash command to reload the configuration in the current session
 - This demonstrates that configuration changes don't apply retroactively to already-running processes
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Section 3: Editing Configuration Files for Services

Tasks Completed:

- Opened nginx configuration file using sudo and nano
- Modified the port number from 81 to 8080
- Changed the root directory to /usr/share/nginx/html/text_ed_lab/
- Started nginx with the modified configuration
- Tested the web server at 127.0.0.1:8080
- Stopped the nginx service

Configuration Changes Made:

File: /etc/nginx/custom_server.conf

Changes:

1. Line 39: Port changed from 81 to 8080
2. Line 47: Root path changed to `/usr/share/nginx/html/text_ed_lab/`

Commands Used:

```
sudo nano -l /etc/nginx/custom_server.conf
```

```
sudo nginx -c custom_server.conf
```

```
sudo pkill nginx
```

Question 6: What is the error message referring to?

Answer: The error message likely refers to:

- Issues with the configuration file syntax
- Missing or incorrect file paths (the `text_ed_lab` directory might not exist)
- Permission issues accessing the specified directory
- Port conflicts if port 8080 is already in use
- The `-l` flag in nano showed line numbers which helped identify the exact location of the error

Question 7: Does the web page appear?

Answer: No, the web page does NOT appear after shutting down nginx.

Explanation:

- Once the nginx service is stopped using `sudo pkill nginx`, the web server no longer runs
- Attempting to access `127.0.0.1:8080` after the server is stopped results in a connection error
- The browser cannot reach the service because there is no server listening on that port
- This confirms that nginx was successfully shut down

Challenge Question

Can you edit the `/etc/nginx/custom_configuration.conf` file with SciTE? Describe the process below.

Answer: Yes, it is possible to edit system files like `/etc/nginx/custom_configuration.conf` with SciTE, but it requires root privileges.

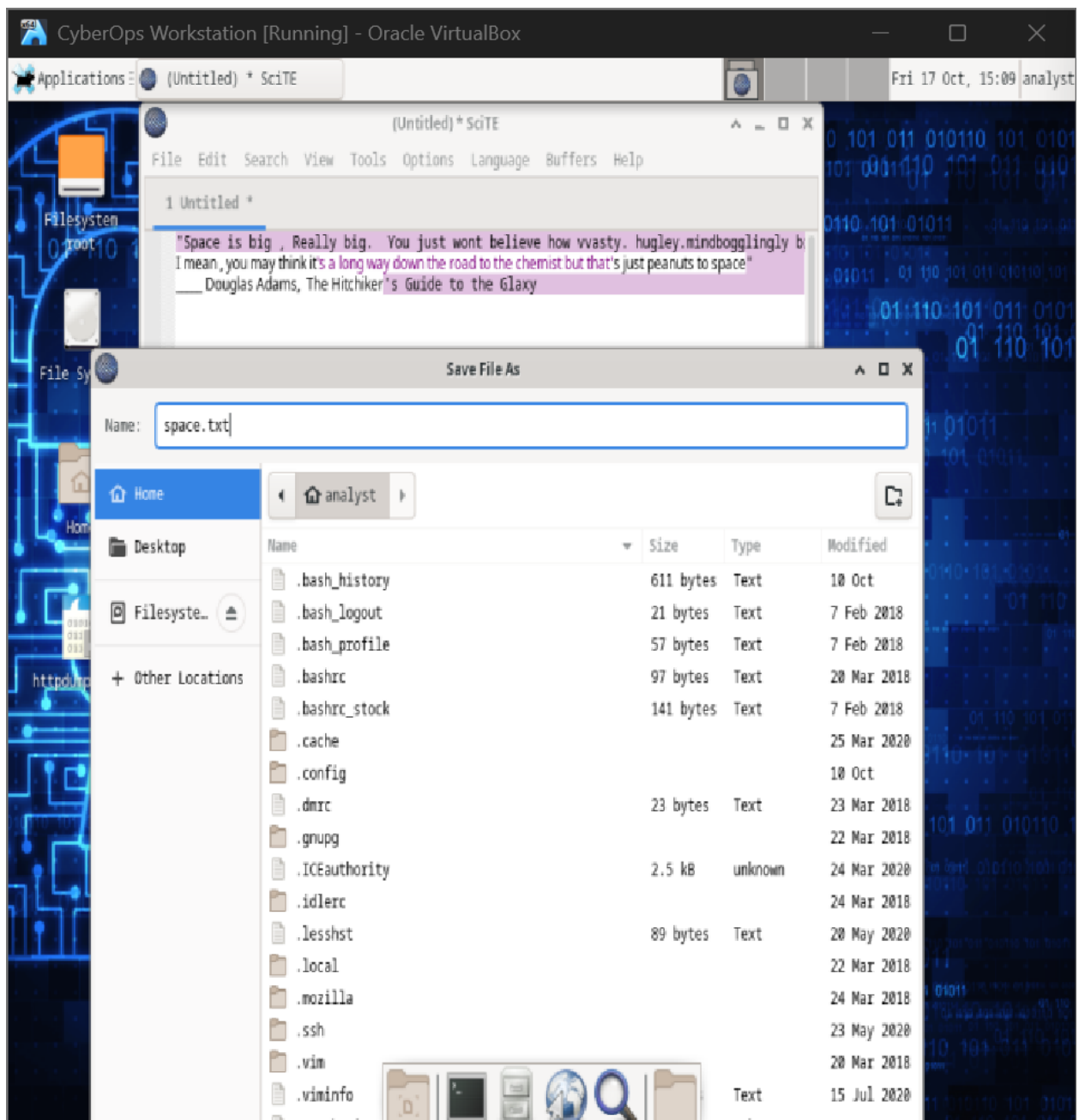
Process:

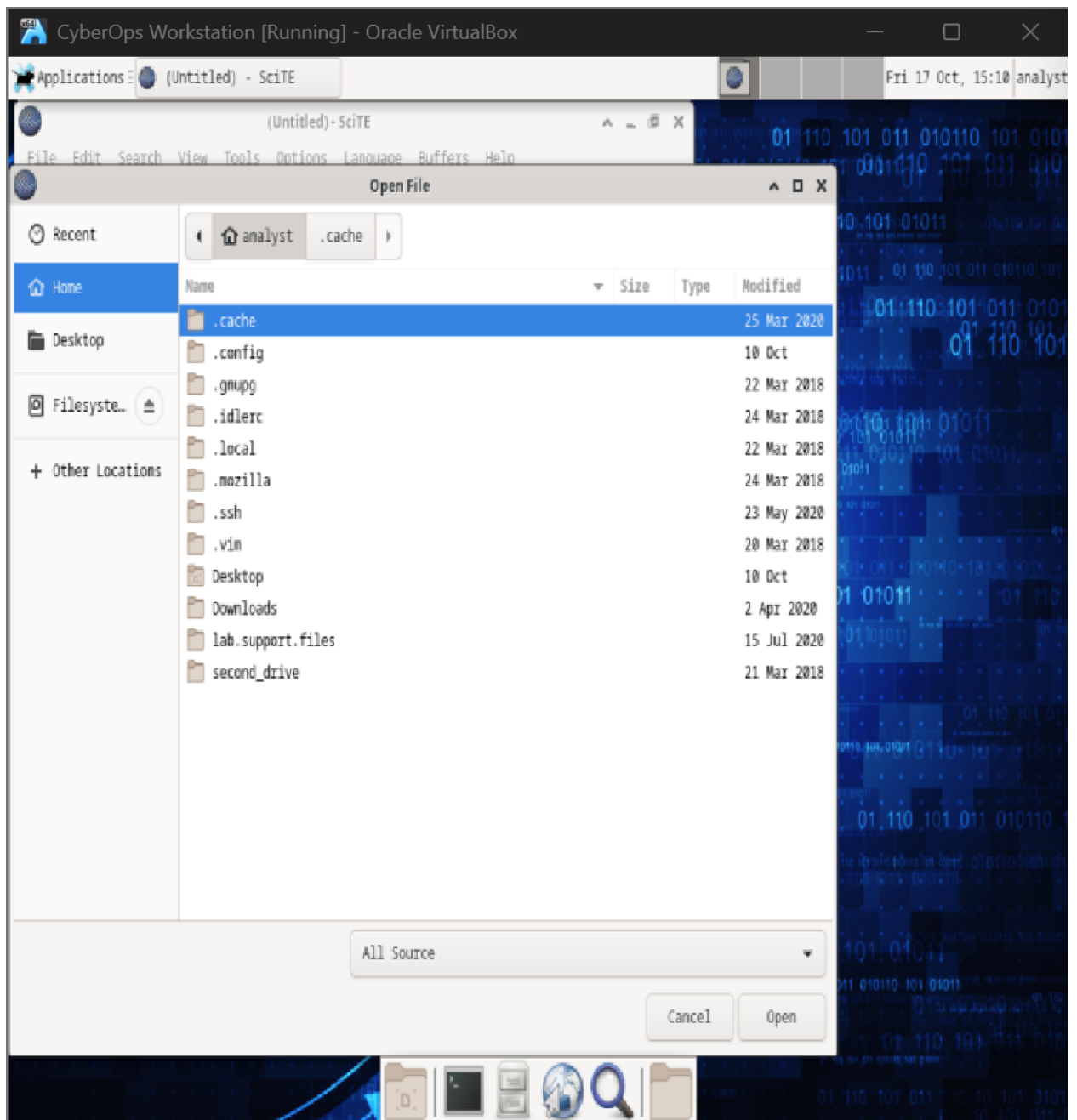
- 1. Open a terminal window**
- 2. Execute the command: `sudo scite /etc/nginx/custom_server.conf`**
- 3. Enter the password for the analyst user when prompted**
- 4. SciTE will open with root privileges, allowing access to the file in `/etc`**
- 5. Make the necessary modifications to the configuration file**
- 6. Save the file using `File > Save`**
- 7. Close SciTE**

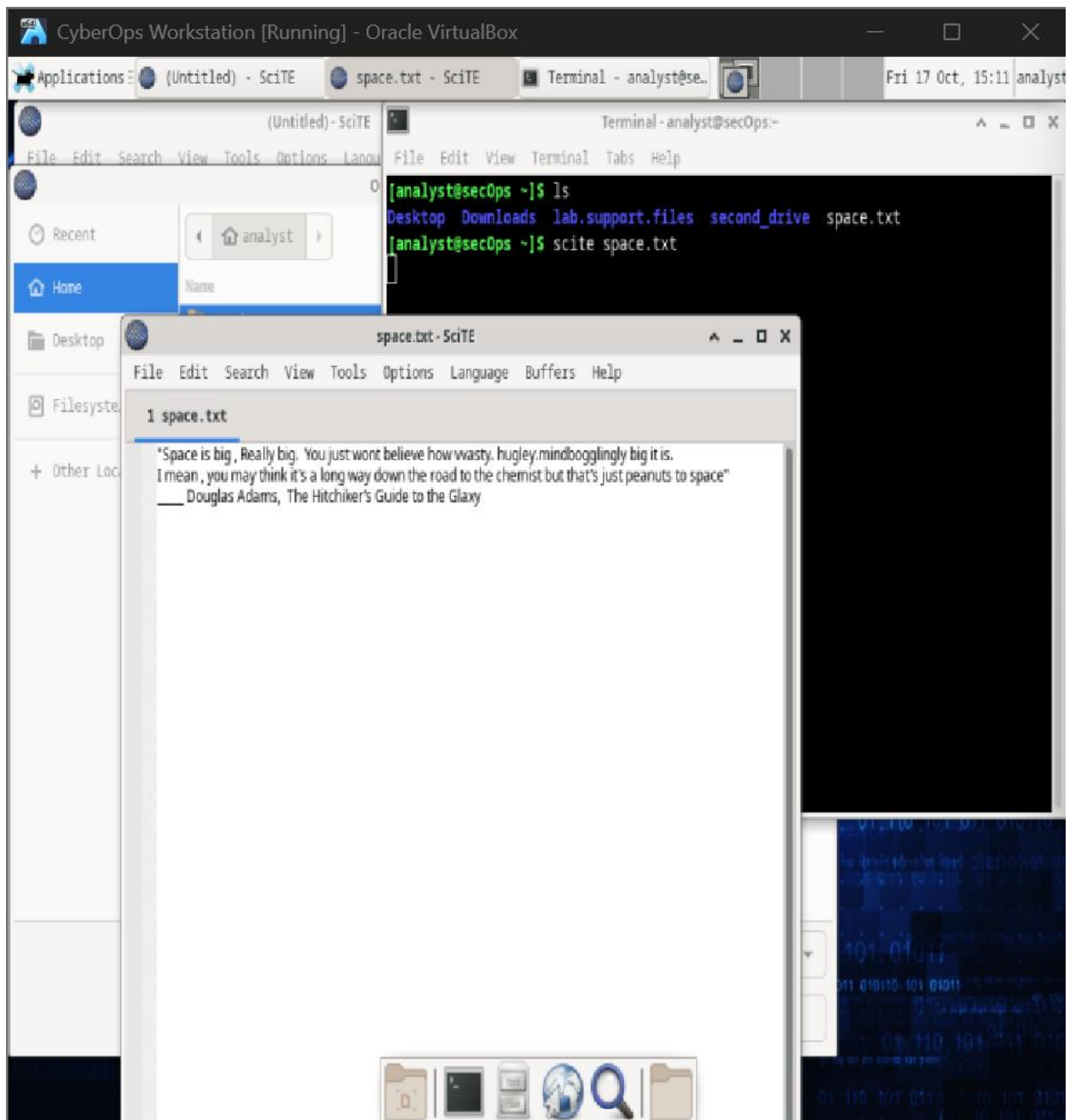
Important Notes:

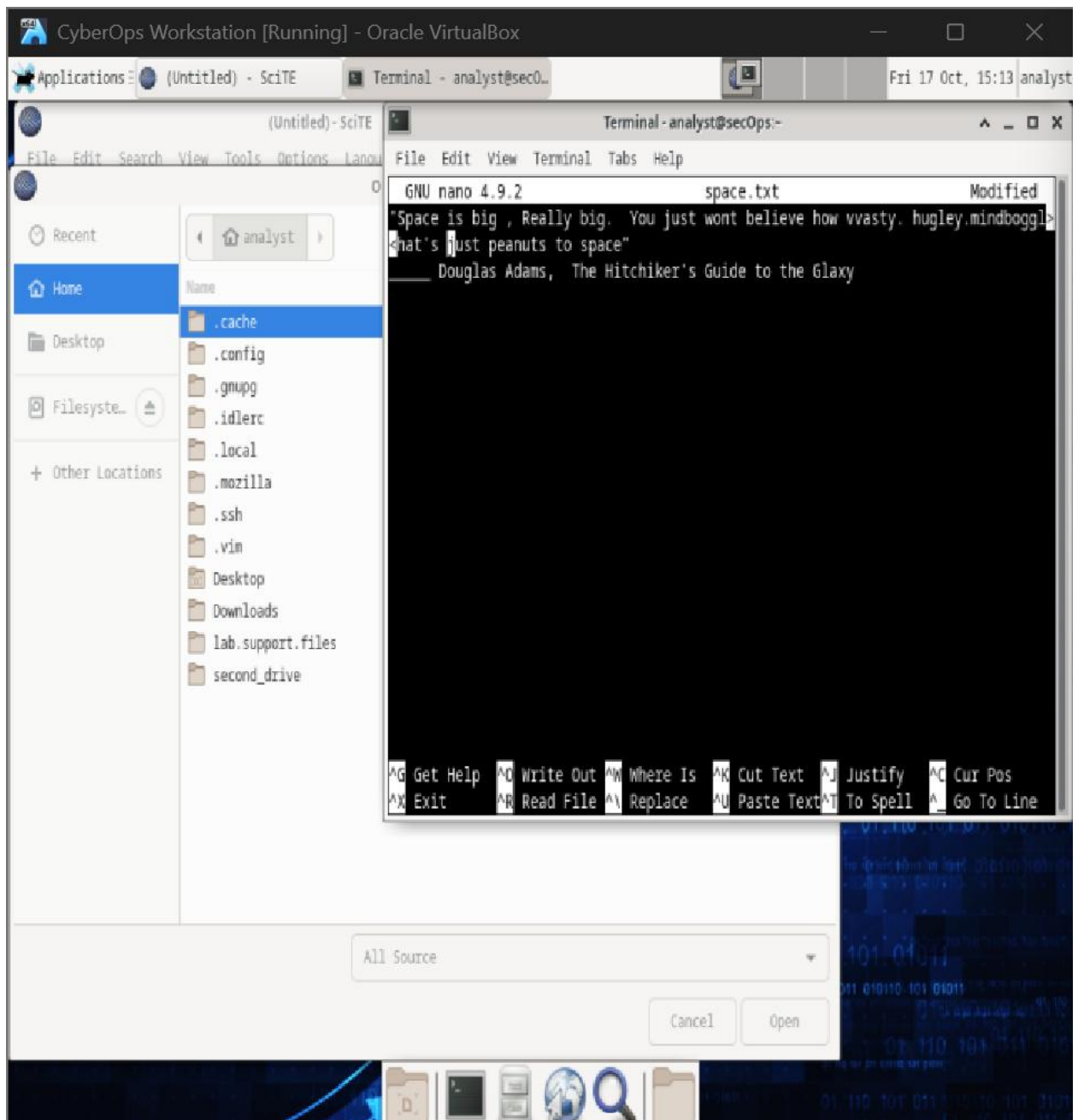
- The `sudo` command grants elevated privileges to SciTE**
- Without `sudo`, SciTE would not have permission to access files in `/etc`**
- This demonstrates the use of privilege escalation for administrative tasks**
- Changes made to system configuration files require `sudo` and affect all users of the system**

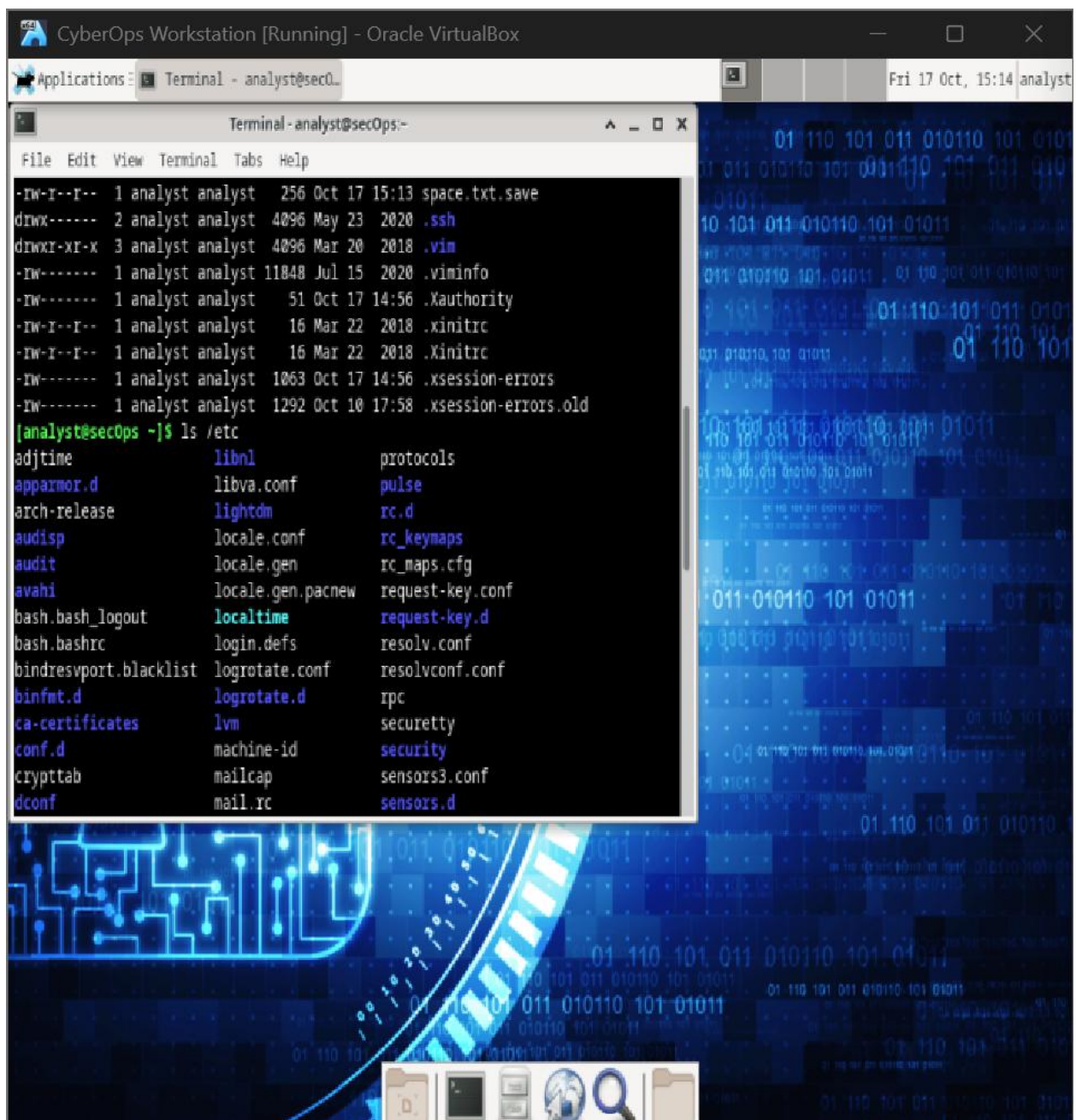
Screenshots:

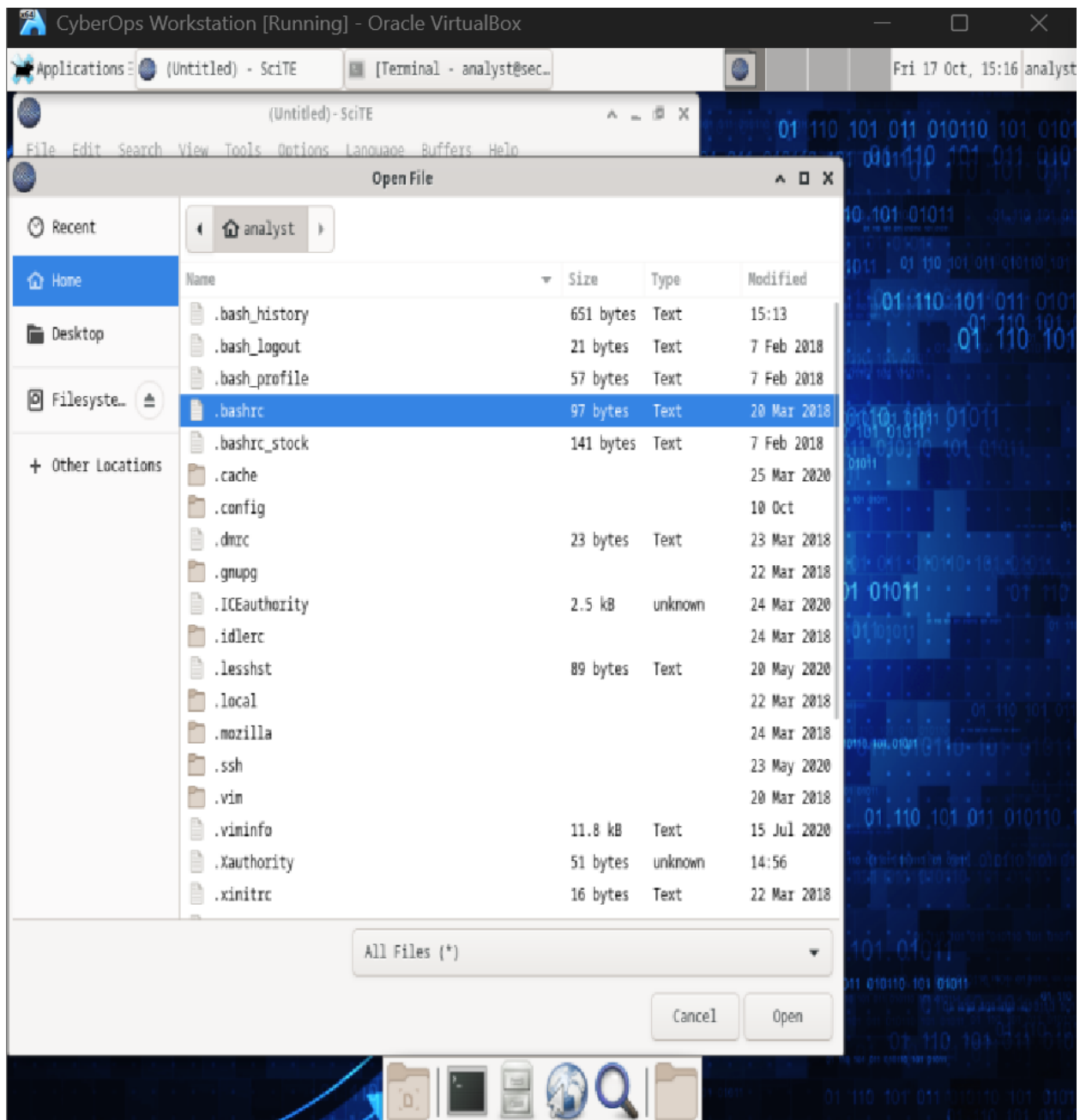


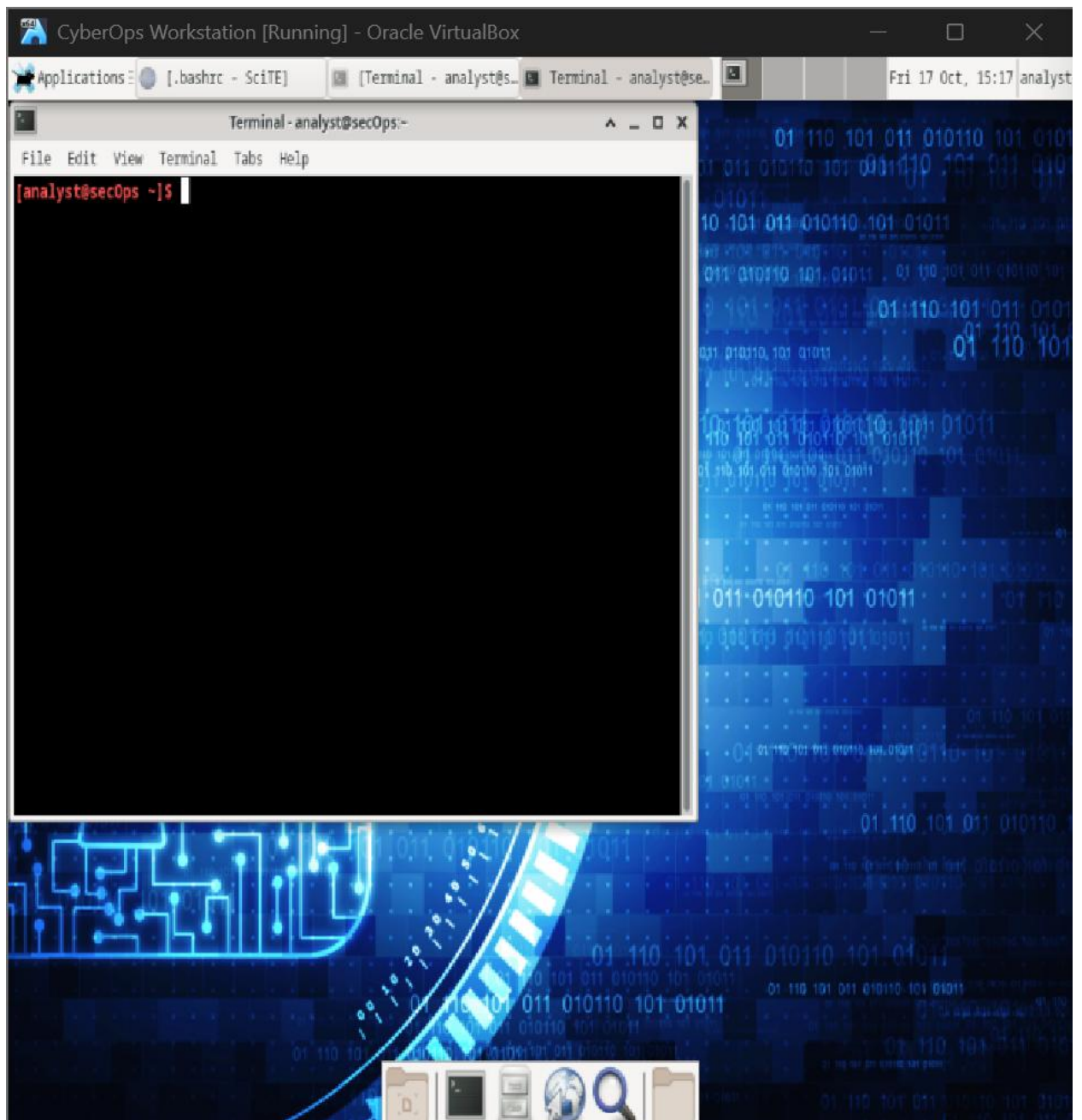


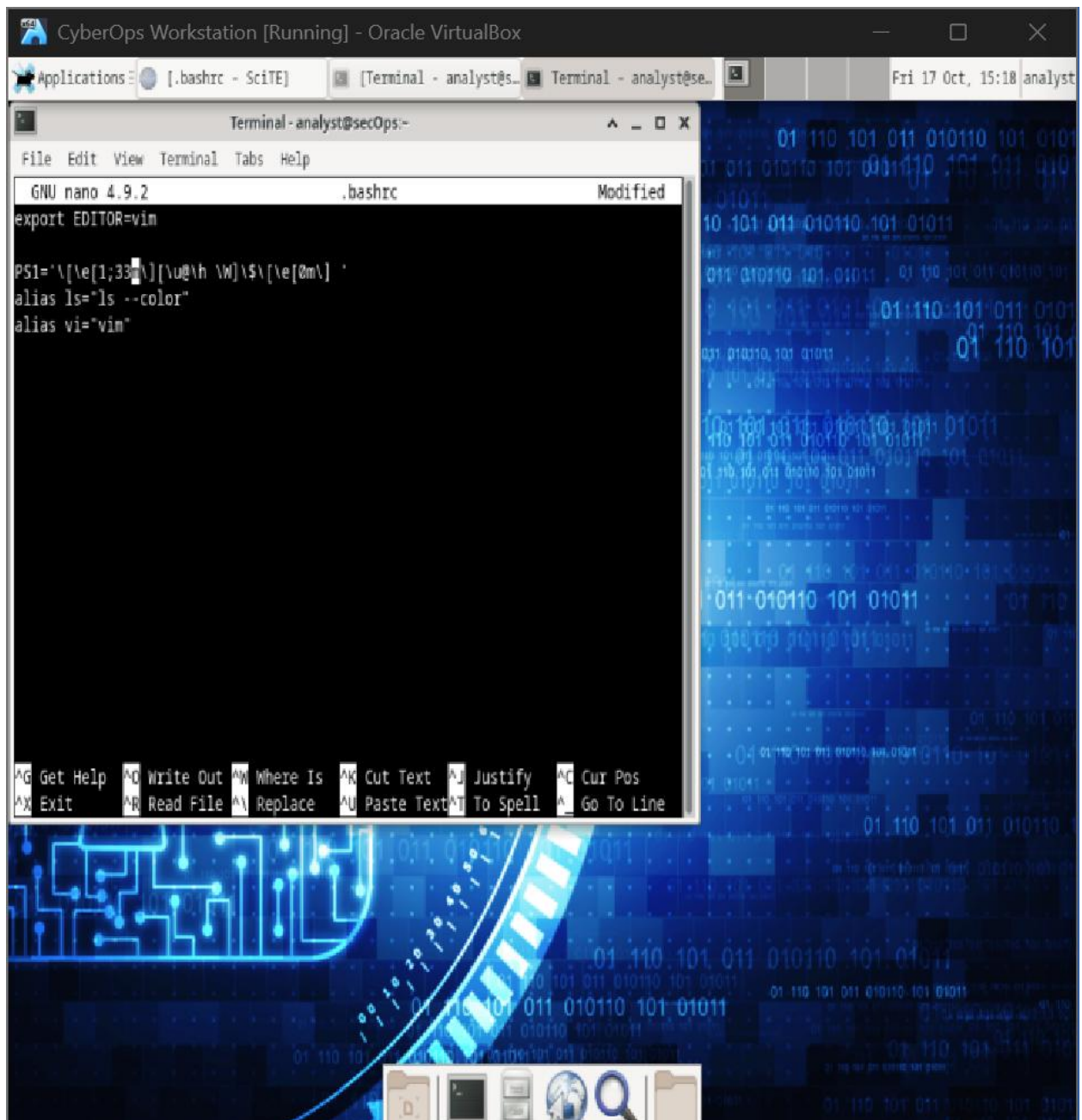


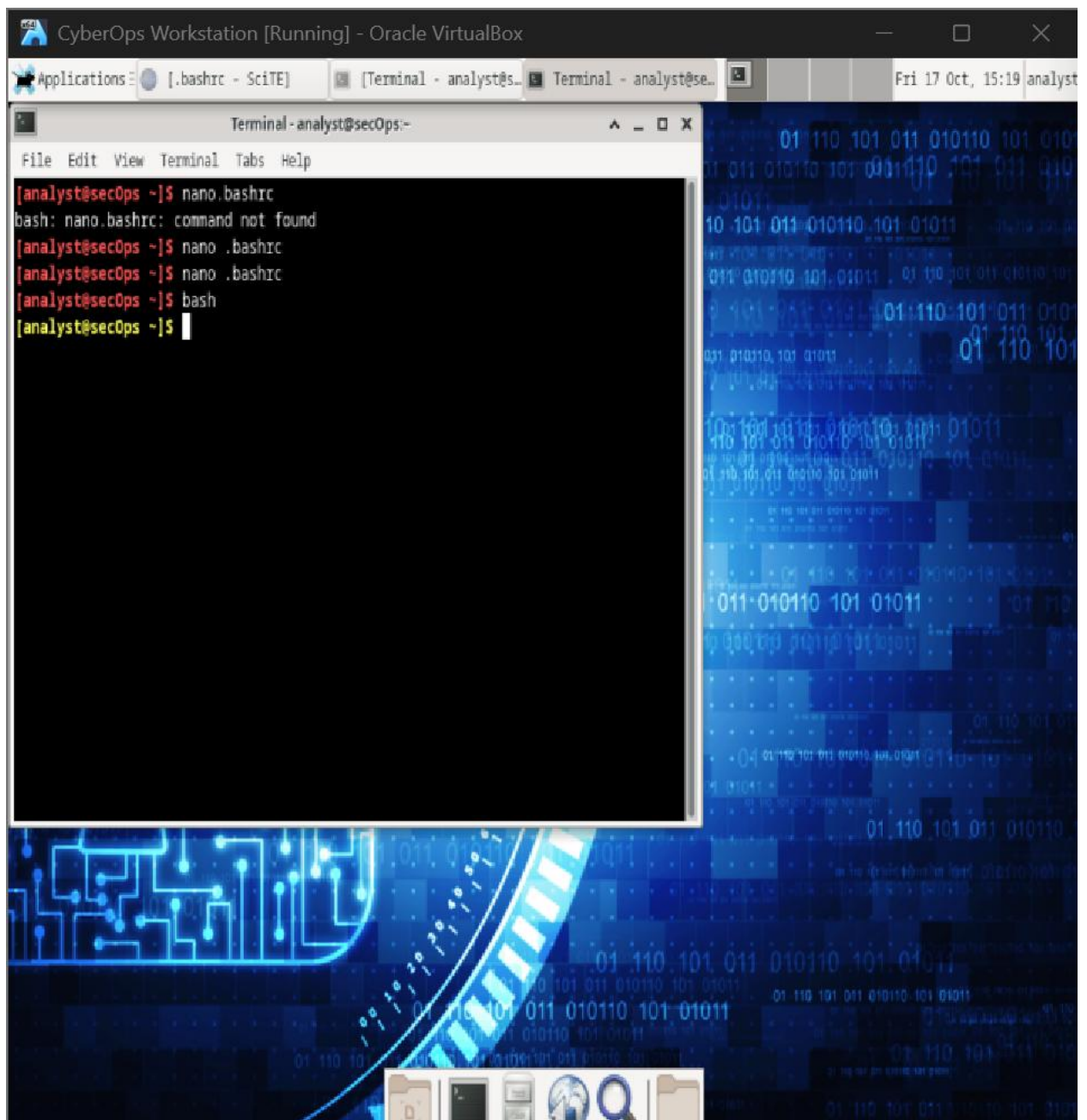


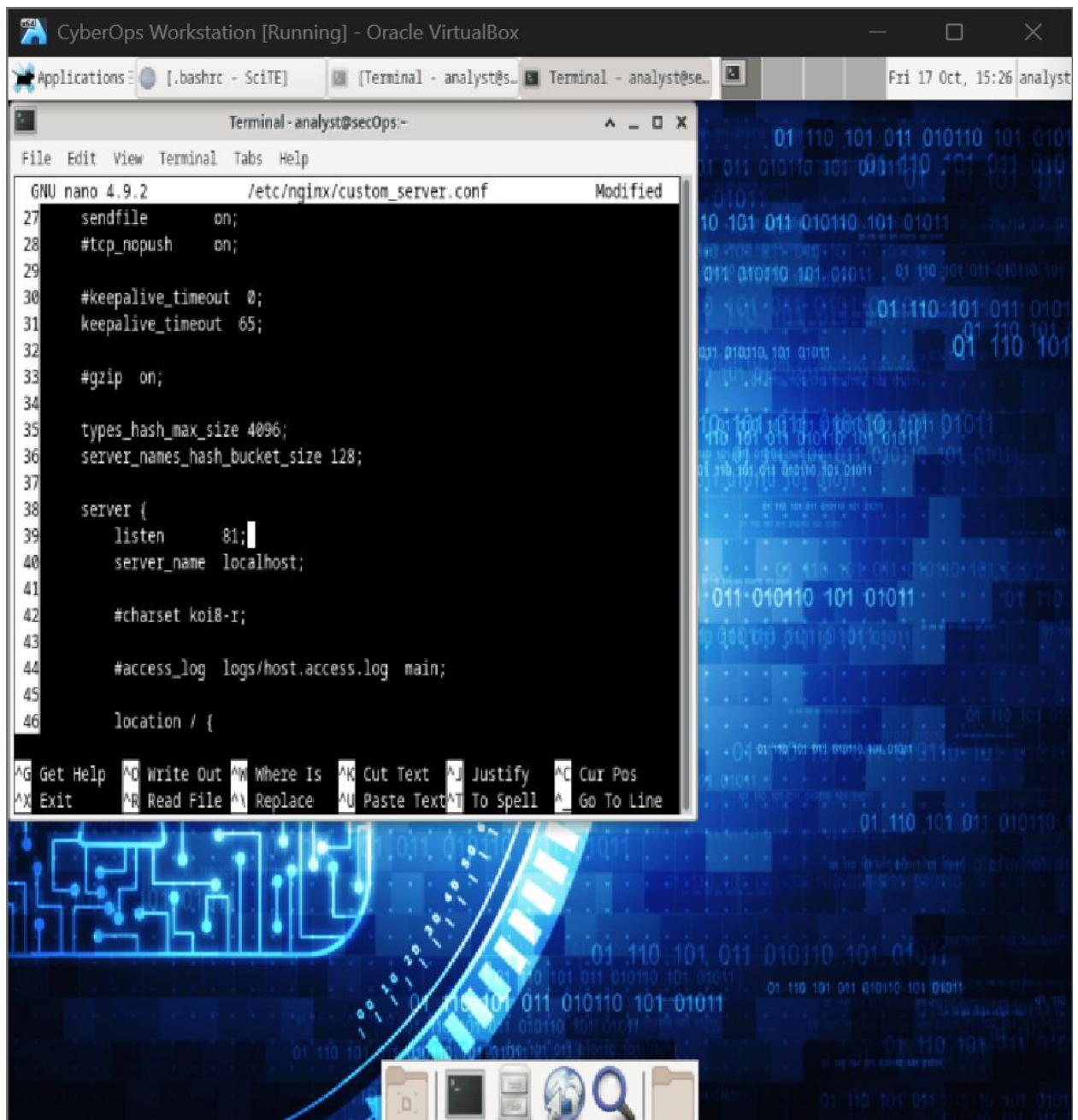


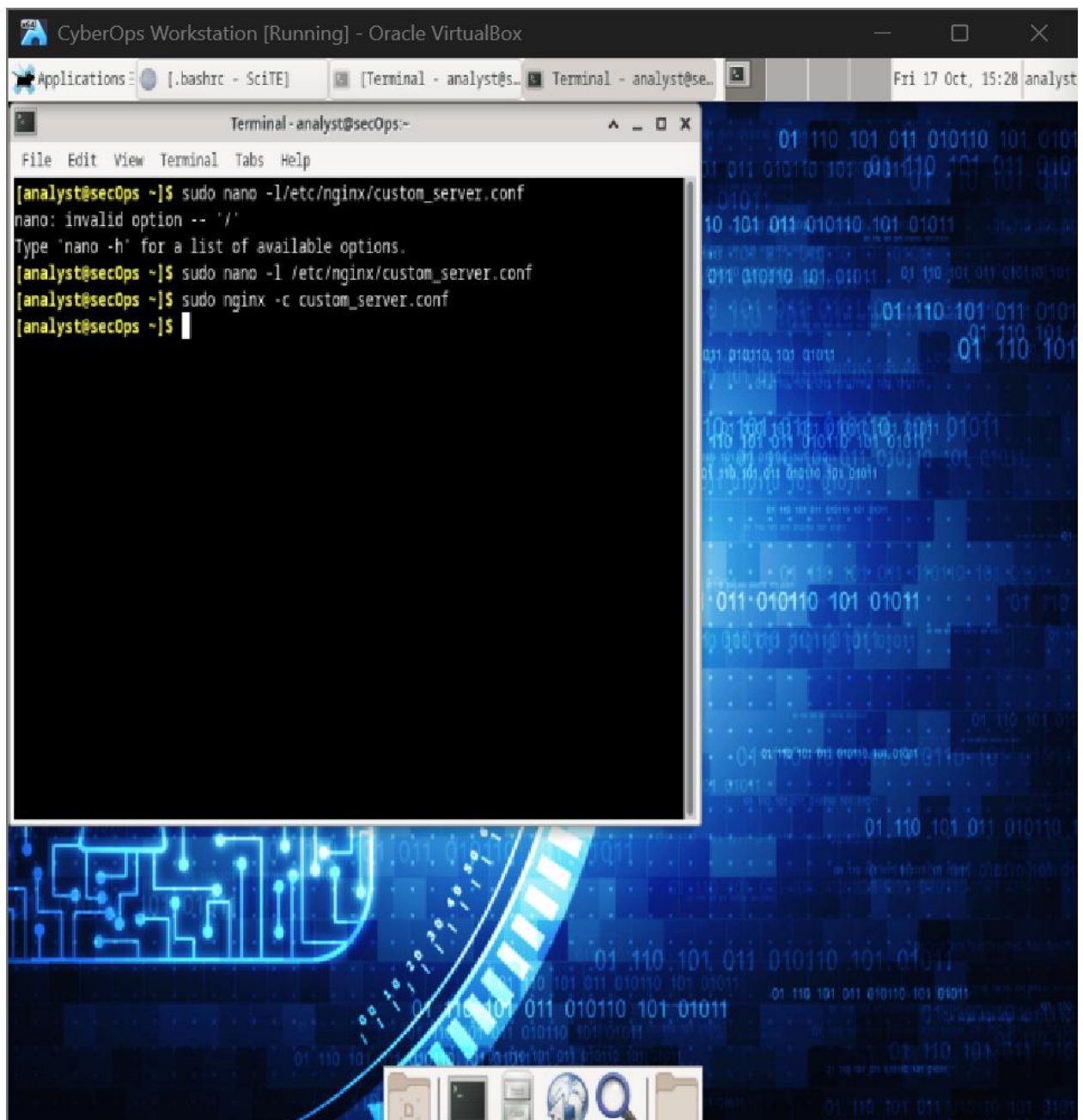


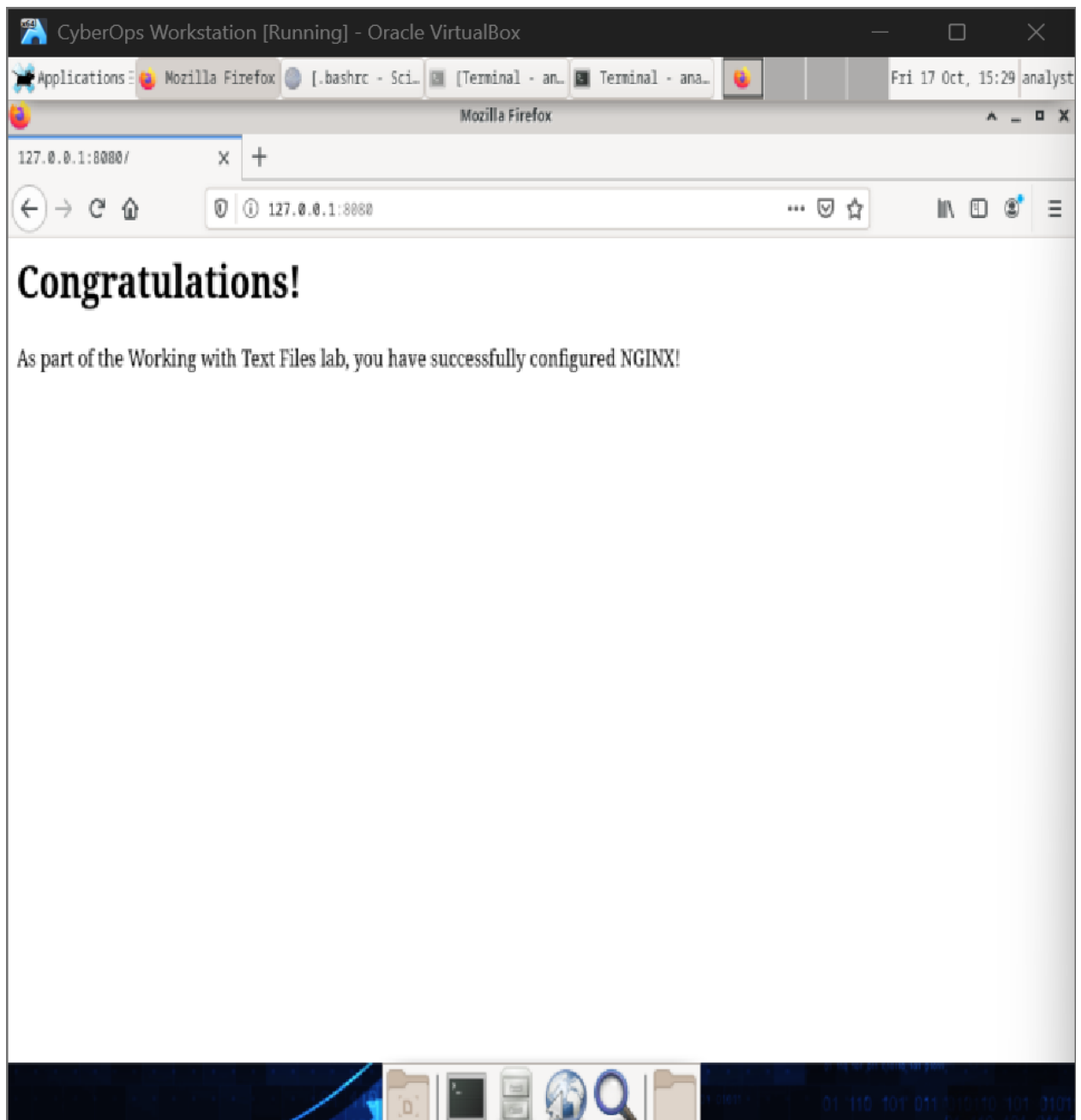


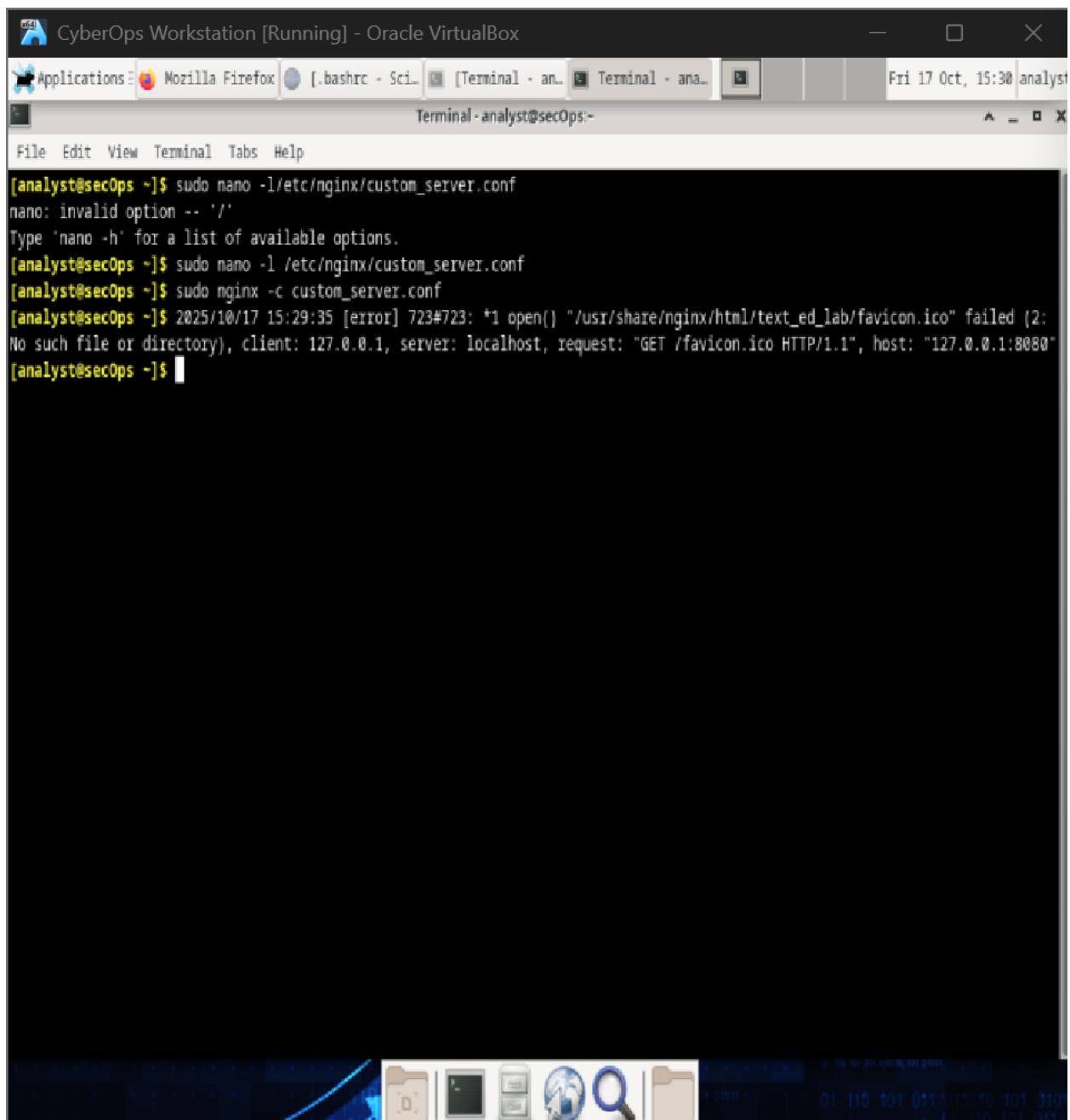


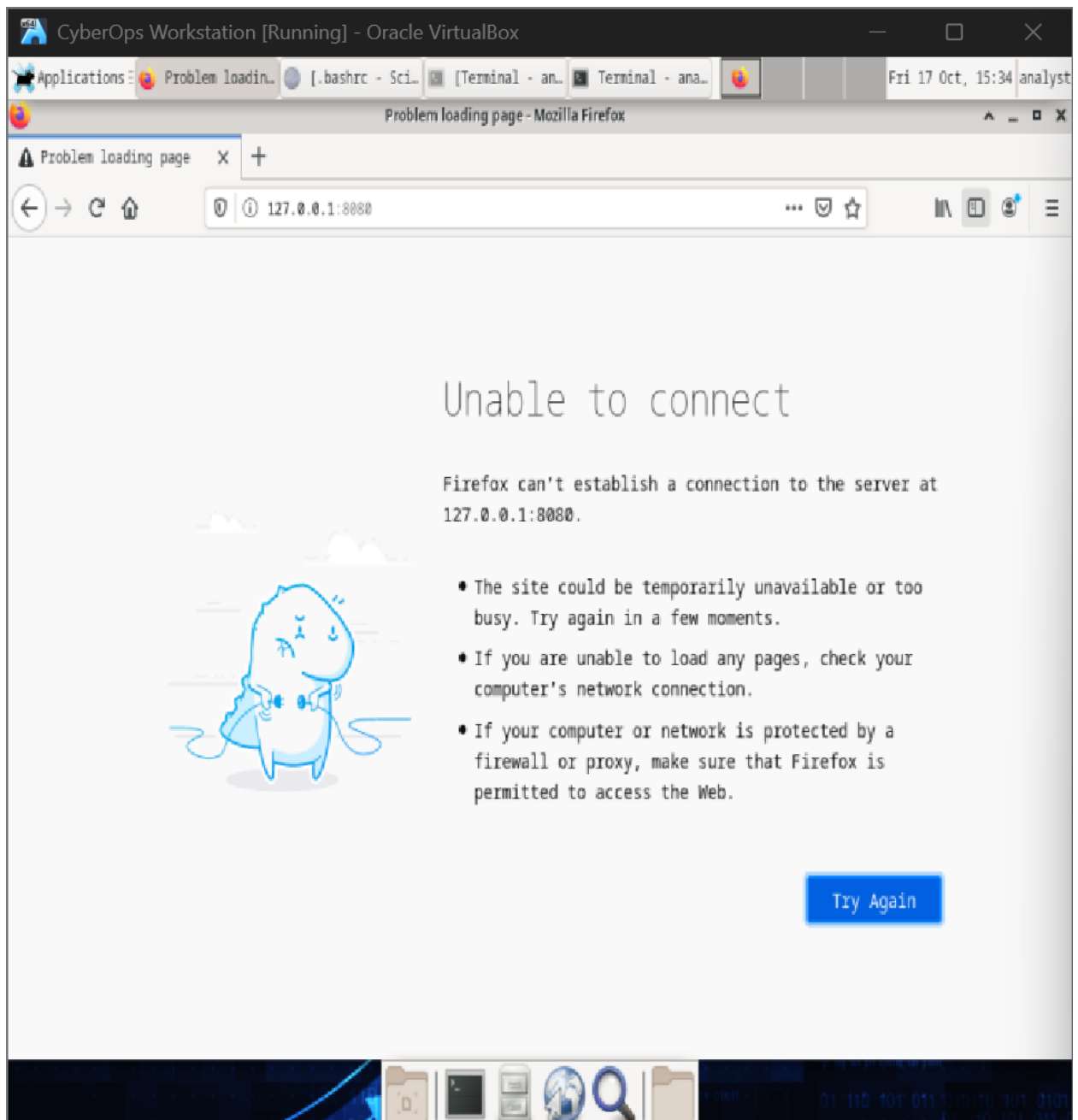


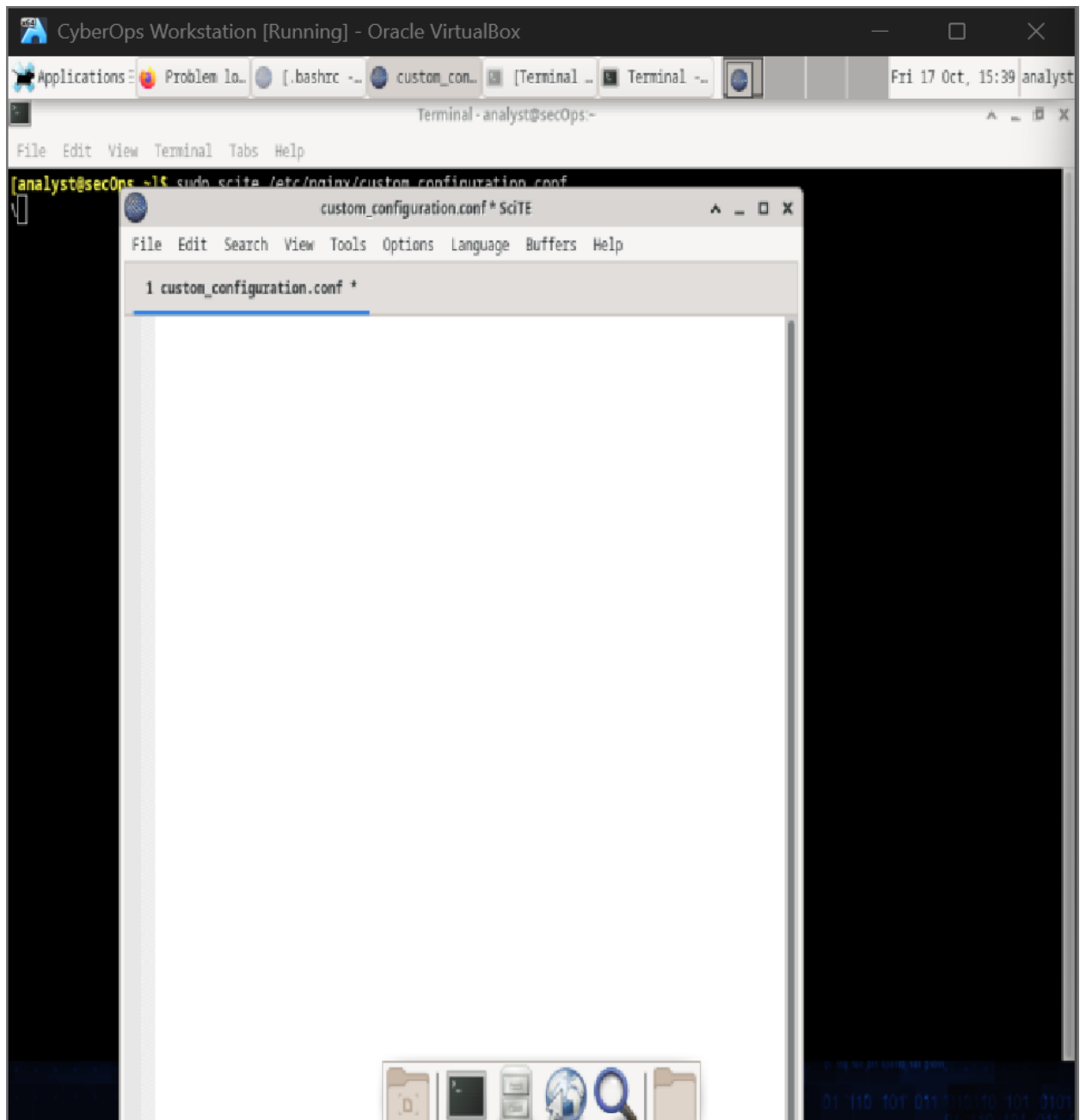












Key Learnings and Conclusions

Important Concepts Learned:

1. Text Editor Types:

- Graphical editors (SciTE) are user-friendly but require GUI

- **Command-line editors (Nano) are essential for remote connections and server administration**

2. Configuration Files:

- **User configs: ~ (home directory)**
- **System configs: /etc directory**
- **Proper permissions are critical for security**

3. File Permissions:

- **Linux uses permissions to control access to files and directories**
- **Understanding permissions prevents accidental system damage**
- **sudo should be used cautiously for administrative tasks**

4. Service Configuration:

- **Services read configuration files on startup**
- **Changes to config files require service restart to take effect**
- **Testing is essential after making configuration changes**

5. Terminal Behavior:

- **Configuration changes don't affect already-running sessions**
- **New terminal sessions load the latest configuration**
- **The bash command can reload configuration in current session**